Damian Sójka

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Research Summary

PhD Candidate at Poznan University of Technology and IDEAS NCBR specializing in **Test-Time Adaptation (TTA)**, **Machine Perception**, and **Robotics**. My research aims to build robust, intelligent perception systems capable of continuous learning and adaptation in dynamic environments. I am the **Principal Investigator (PI)** of a competitive, 2-year PRELUDIUM grant funded by the National Science Centre, and my background includes **embedded R&D for bionic hand prostheses** at Aether Biomedical. Currently, as a **MARS Fellow** at the Cambridge AI Safety Hub, I benchmark **LLM capabilities** for agentic AI development. My research is published in premier venues including **ICRA**, **BMVC**, and specialized workshops at **ICML** and **ECCV**, and was recognized with an Innovation Award at **ICCV** workshop.

Professional Experience

Dec 2022 – present
PhD Researcher, IDEAS NCBR (Continual Machine Learning Group)
- Conducting research on TTA and CV at a leading Polish AI R&D center.

Dec 2025 – present
MARS Fellow, Cambridge AI Safety Hub
- **Benchmarking Agentic AI:** Collaborating with Arcadia Impact to develop evaluations for agentic LLM capabilities in AI R&D contexts.
- **Framework Development:** Engineering benchmarks within the UK AI Safety Institute's (**AISI**) *Inspect Evals* open-source framework.
- **Automated Validation:** Developing an LLM-based scanner for transcript analysis to detect benchmark validity violations.

Sep 2024 – Nov 2024
Research Intern, ENSTA Paris
- **TTA for Depth Estimation:** Worked on improving reliability of depth estimation under distribution shifts via TTA with Profs. David Filliat and Antoine Manzanera.

Jul 2021 – Nov 2022
Embedded Software Developer, Aether Biomedical
- Led R&D for bionic hand software, including real-time logging system.

2019 – 2020
Industrial Experience (Farmutil HS, Qubiqa)
- Assembly of control cabinets and maintenance of industrial production lines.

Education

Jan 2023 – now
PhD in Deep Learning, Poznan University of Technology.
- My research is focused on TTA and Machine Perception.
- Supervisors: Prof. Piotr Skrzypczyński and Prof. Michał Nowicki.

Mar 2021 – Sep 2022
MSc in Automatic Control and Robotics, Poznan University of Technology.
- MSc thesis: *Triplet Loss in Haptic Localization of a Walking Robot*.
- Member of a student team building autonomous race car to compete in Formula Student events.
- Rector's Scholarship for high-achieving students.

Oct 2017 – Feb 2021
BSc in Mechatronics, Poznan University of Technology.
- BSc thesis: *Control of Logistics Tractor Using Artificial Intelligence Methods*.
- Rector's Scholarship for high-achieving students.

Papers

1. S. Cygert, **D. Sójka**, T. Trzciński, and B. Twardowski, "Realistic evaluation of test-time adaptation algorithms: Unsupervised hyperparameter selection", in *Proceedings of the 21st International Conference on Computer Vision Theory and Applications (VISAPP)*, 2026. URL: https://arxiv.org/pdf/2407.14231.
2. **D. Sójka**, M. Masana, B. Twardowski, and S. Cygert, "Adaptive monocular depth estimation with masked image consistency", in *Second Workshop on Test-Time Adaptation at ICML*, 2025. URL: https://openreview.net/pdf?id=qqTYN7O5w5.
3. **D. Sójka**, M. Masana, B. Twardowski, and S. Cygert, "Intransigent teachers guide better test-time adaptation students", in *Out-Of-Distribution Generalization in Computer Vision Workshop at European Conference on Computer Vision (ECCV)*, 2024. URL: https://openreview.net/pdf?id=Chq4OQ3p18.
4. **D. Sójka**, M. R. Nowicki, and P. Skrzypczyński, "Triplet loss-based metric learning for haptic-only robot localization", in *Progress in Polish Artificial Intelligence Research 5 : Proceedings of the 5th Polish Conference on Artificial Intelligence (PP-RAI)*, 2024. URL: https://pages.mini.pw.edu.pl/~estatic/pliki/PP-RAI_2024_proceedings.pdf#page=357.
5. **D. Sójka**, B. Twardowski, T. Trzcinski, and S. Cygert, "Ar-tta: A simple method for real-world continual test-time adaptation", in *35th British Machine Vision Conference (BMVC)*, BMVA, 2024. URL: https://arxiv.org/pdf/2309.10109.
6. **D. Sójka**, M. R. Nowicki, and P. Skrzypczyński, "Learning an efficient terrain representation for haptic localization of a legged robot", in *IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, 2023. URL: https://arxiv.org/pdf/2209.15135.

Other Achievements

Awards and Grants

2024 Polish National Science Centre PRELUDIUM grant for a research project "Reliable and Efficient Real-World Test-Time Adaptation".

2023 Innovation award and 3rd place in Continual Test Time Adaptation for Semantic Segmentation challenge organized within Visual Continual Learning workshop at ICCV 2023.

Presentations

2025 **Ellis Doctoral Symposium on Robust AI.** Presented a poster "Adaptive monocular depth estimation with masked image consistency".

2024 **International Computer Vision Summer School (ICVSS).** Presented a poster "AR-TTA: A Simple Method for Real-World Continual Test-Time Adaptation".

2023 **ML in PL Conference.** Presented a poster "AR-TTA: A Simple Method for Real-World Continual Test-Time Adaptation".

Skills

Machine Learning Expert-level PyTorch, TensorFlow, WandB, Scikit-learn. Experience in TTA, OOD generalization, Computer Vision, and LLM benchmarking (Inspect Evals).

Languages Python (Proficient), C (Proficient), C++ (Basic), LaTeX, Bash. Fluency in English and Polish.

Robotics & Tools Embedded Software Development, ROS2, Docker, OpenCV, SLURM, Git, Linux.

Scientific Writing Paper and grant proposal writing (Principal Investigator), peer-reviewing, and presenting at international conferences (ICRA, BMVC, ICCV).